

BAYKIT, Russian Federation

WMO: 238910

Lat: 61.6801N

Lon: 96.3630E

Elev: 860

StdP: 14.25

Time Zone: 7.00 (E07)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -53.0	(c) -49.0	(d) -56.7	(e) 0.2	(f) -53.1	(g) -52.5	(h) 0.3	(i) -49.0	(j) 11.5	(k) 3.4	(l) 9.3	(m) -3.7	(n) 0.0	(o) 350	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 20.9	(c) 83.9	(d) 64.6	(e) 80.0	(f) 62.0	(g) 75.8	(h) 60.6	(i) 66.6	(j) 79.0	(k) 64.6	(l) 75.4	(m) 62.6	(n) 71.8	(o) 4.2	(p) 140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.6	87.7	70.0	60.6	81.4	67.7	58.7	76.0	66.9	31.7	78.6	30.2	76.0	28.6	71.5	73.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 12.6	(b) 10.2	(c) 8.6	(e) -52.9	(f) 88.6	(g) 7.1	(h) 4.3	(i) -58.0	(j) 91.7	(k) -62.2	(l) 94.2	(m) -66.1	(n) 96.6	(o) -71.3	(p) 99.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.7	-22.2	-11.1	11.4	28.5	40.1	58.3	62.5	55.9	42.1	25.4	-3.7	-16.8	
	DBStd	31.68	14.74	15.37	13.36	9.63	7.67	8.10	6.09	6.30	7.33	11.28	18.44	19.49	
	HDD50	10856	2237	1712	1197	646	323	23	1	18	256	762	1611	2070	
	HDD65	15507	2702	2132	1662	1096	772	226	118	290	686	1227	2061	2535	
	CDD50	897	0	0	0	0	15	272	390	200	20	0	0	0	
	CDD65	73	0	0	0	0	0	25	42	6	0	0	0	0	
	CDH74	1210	0	0	0	0	5	528	570	104	3	0	0	0	
	CDH80	321	0	0	0	0	0	158	150	13	0	0	0	0	
Wind	WSAvg	2.8	1.1	1.7	3.2	4.2	4.5	3.7	3.6	3.1	3.0	2.7	1.7	1.1	
Precipitation	PrecAvg	20.1	1.1	0.8	0.8	1.1	1.6	2.0	1.9	3.3	2.2	2.1	1.8	1.3	
	PrecMax	25.4	2.0	2.0	1.7	2.0	3.6	3.7	3.3	5.9	4.0	4.3	3.0	2.4	
	PrecMin	12.8	0.4	0.2	0.0	0.3	0.2	0.5	0.3	0.6	0.5	1.0	0.4	0.7	
	PrecStd	3.0	0.4	0.4	0.4	0.4	0.7	0.8	0.9	1.5	0.9	0.9	0.6	0.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.3	27.0	43.3	56.8	73.0	89.6	87.8	81.9	71.4	49.6	34.7	24.3	
		MCWB	13.3	25.4	33.9	43.7	52.7	66.4	66.6	63.3	57.1	45.7	32.6	23.2	
	2%	DB	10.4	22.0	38.2	51.0	67.1	84.7	84.3	77.1	64.6	45.0	32.0	21.8	
		MCWB	9.3	20.6	31.7	40.3	51.5	63.5	65.8	60.4	53.4	41.3	30.2	20.7	
	5%	DB	4.3	16.4	33.9	46.7	60.9	80.3	80.8	72.2	58.5	41.3	29.0	17.3	
		MCWB	3.4	14.9	28.9	37.6	47.6	61.2	63.0	59.9	50.5	38.8	27.3	16.4	
	10%	DB	-0.9	11.8	30.1	42.9	54.8	75.8	76.5	68.0	53.6	38.0	24.4	12.0	
		MCWB	-1.6	10.7	26.5	35.1	44.2	59.8	61.8	58.6	48.1	35.8	23.1	11.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.7	25.6	35.1	44.7	55.6	68.6	70.6	66.2	58.7	46.8	33.0	23.4
			MCDB	14.2	26.9	41.1	55.0	69.6	85.1	82.8	77.0	66.5	49.6	34.2	24.0
2%		WB	9.5	20.6	32.2	41.1	52.2	65.4	67.5	63.6	55.5	42.0	30.5	20.5	
		MCDB	10.5	22.1	37.3	49.9	65.2	80.1	78.8	71.8	62.7	44.5	31.9	21.6	
5%		WB	3.6	15.0	29.4	38.4	48.7	63.0	65.7	61.4	52.0	38.9	27.3	16.5	
		MCDB	4.4	16.3	33.6	44.8	59.5	76.4	76.6	68.2	56.4	40.8	29.0	17.3	
10%		WB	-1.5	10.7	26.4	36.2	45.3	60.8	63.9	59.4	48.7	36.0	22.9	11.2	
		MCDB	-0.9	12.0	30.1	41.7	53.5	72.8	73.2	66.6	53.0	37.7	24.3	12.0	
Mean Daily Temperature Range		5% DB	MDBR	12.3	18.2	22.5	19.1	18.4	22.9	20.9	18.3	15.6	10.2	13.1	12.1
			MCDBR	15.7	17.0	22.7	23.2	30.2	32.9	29.4	26.6	26.6	13.6	11.3	13.3
	MCWBR		15.3	15.1	16.9	14.0	16.2	13.8	12.8	12.5	14.8	10.0	10.6	12.9	
	5% WB	MCDBR	15.6	15.8	21.6	21.0	28.0	28.9	24.5	22.2	21.7	11.3	11.2	13.3	
		MCWBR	15.2	14.8	16.3	13.8	16.0	13.2	12.3	12.2	13.0	9.0	10.6	12.9	
Clear-Sky Solar Irradiance	taub	0.250	0.289	0.318	0.349	0.364	0.387	0.427	0.417	0.341	0.301	0.237	0.213		
	taud	1.919	1.989	2.017	2.038	2.233	2.369	2.246	2.220	2.416	2.224	1.969	1.862		
	Ebn at Noon	127	203	245	262	270	266	250	240	244	204	133	84		
	Edh at Noon	14	25	36	43	38	35	38	36	24	20	12	8		
All-Sky Solar Radiation	RadAvg	110	400	878	1370	1588	1771	1602	1138	691	307	138	55		
	RadStd	10	28	62	119	121	173	143	98	104	43	19	7		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page